

proxy_socks5_challenge.sh

Revision: 1 **Last modified:** 2026-07-01T00:00:00Z **Authority:** Helix Constitution §11.4.27 (Challenges), §11.4.69 (sink-side positive evidence), §11.4.1 (no false-FAIL), §11.4.68 (no fail-open), §11.4.2/§11.4.5 (captured evidence)

Overview

Anti-bluff Challenge that proves the **live SOCKS5 proxy** (default socks5://localhost:51080) actually forwards a real end-user journey. It drives two real sub-probes through `curl --proxy socks5://...` — a plain-HTTP GET to a 204 endpoint and an HTTPS GET — and asserts the through-proxy `%{http_code}` matches the expected code, cross-checked against a **direct** fetch of the same URL. Response headers and per-probe verdicts are captured to an evidence file.

Prerequisites

- The base proxy stack UP with SOCKS5 listening on 51080.
- `curl` built with SOCKS5 support, `bash`, `awk`, `grep`; `tests/lib/evidence.sh` present (sourced).
- READ-ONLY: plain proxy client only — never touches any container.

Usage

```
bash challenges/scripts/proxy_socks5_challenge.sh
# override target / evidence dir:
SOCKS5_PROXY=socks5://localhost:51080 \
CHALLENGE_EVIDENCE_DIR=/path/to/evidence \
  bash challenges/scripts/proxy_socks5_challenge.sh
```

Environment: `SOCKS5_PROXY` (default socks5://localhost:51080), `SOCKS5_PORT` (default 51080), `CHALLENGE_EVIDENCE_DIR` (default qa-results/challenges/<run-ts>), `CURL_MAX_TIME` (default 20).

Exit codes

Code	Meaning
0	PASS — ≥ 1 sub-probe proved forwarding and none FAILED.
1	FAIL — a real proxy defect (proxy missed but the same URL was reachable directly).
3	SKIP — honest non-applicable: no endpoint reachable (third-party/network outage, §11.4.3).

Anti-bluff decision

Identical `proxy_conn_verdict` discipline as the HTTP sibling, applied to the SOCKS5 port: `proxy-hits-expected` → PASS; `proxy-miss` + `direct-hit` → FAIL; `both-miss` → SKIP. Aggregate: FAIL if any FAILs; PASS if ≥ 1 PASSes and none FAIL; else SKIP.

Edge cases

- SOCKS5 to plain HTTP shows no `Via:` header (opaque TCP relay); the HTTP status code plus the direct cross-check is the functional evidence.
- A site outage SKIPS the affected sub-probe rather than false-FAILing a healthy proxy.

Outputs

`<evidence-dir>/socks5/socks5_evidence.txt` plus per-probe raw header dumps.

Related scripts

- `proxy_forward_http_challenge.sh` — the HTTP sibling.
- `proxy_cache_challenge.sh` — Squid cache-HIT proof.
- `run_proxy_challenges.sh` — the bank runner.
- `tests/lib/evidence.sh` — the sourced anti-bluff helper library.

Last verified: 2026-07-01 (live run: PASS, 204 + HTTP/2 200 through SOCKS5 captured).